Module :- 5 Special purpose diode

(*) zener diode

Defination :- A zener diode is a type of diode that allows current to flow not only from it's anode to cathode, but also in the reverse direction, when the zener voltage is reached.

> zener diode have a highly doped p-n junction.
> normally diode break down with a reverse voltage, But the zener diode is designed to operate in a break down region.

> However constructing the zener diode, the reverse breakdown voltage is adjusted precisely between 3V to 200V
> it's application are based on this principle hence zener diode is also called as a breakdown diode.

> The dopping level of the impurity added to manufacture the zener diode is controlled in order to adjust the breakdown voltage.

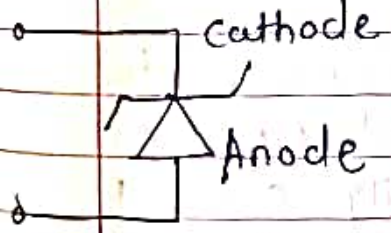


fig a) Symbol of zener diode

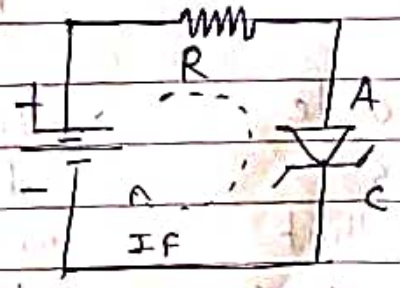


fig b) Forward biasing of zener diode

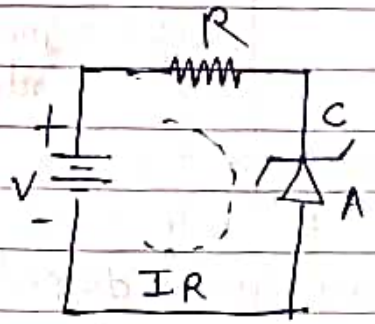
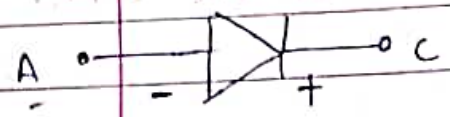


fig c) Reverse biasing of zener diode.

P-N-junction

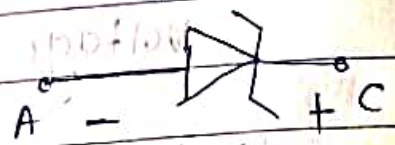
Reverse supply is applied.



That time P-N junction diode is heat & break

zener diode

Reverse supply is applied



That time zener diode does not heat & break.

> Zener diode is special type of P-N junction diode. & it's speaciality is heavily doped diode.

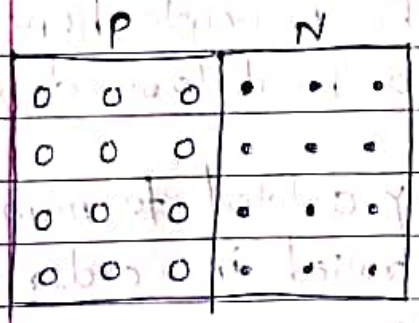


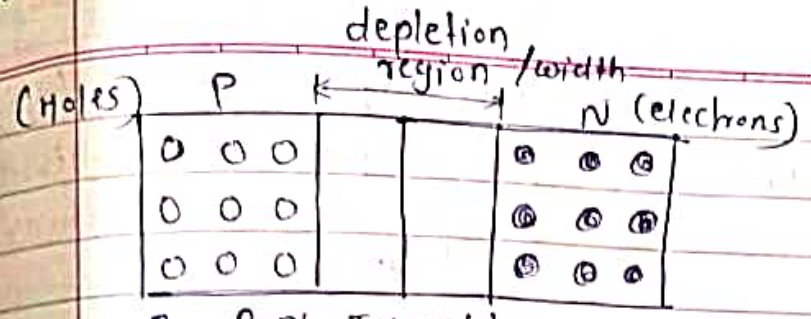
fig a) PN junction diode

fig b) Zener diode

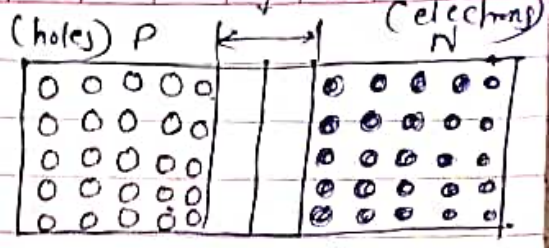
> In a P-N junction diode the holes & electrons are lightly dopped for that reason the depletion region Thick

> In a Zener diode the holes and electrons are heavily dopped for that reason the depletion region is Thin.

Si = 0.7 V
Ge = 0.3 V

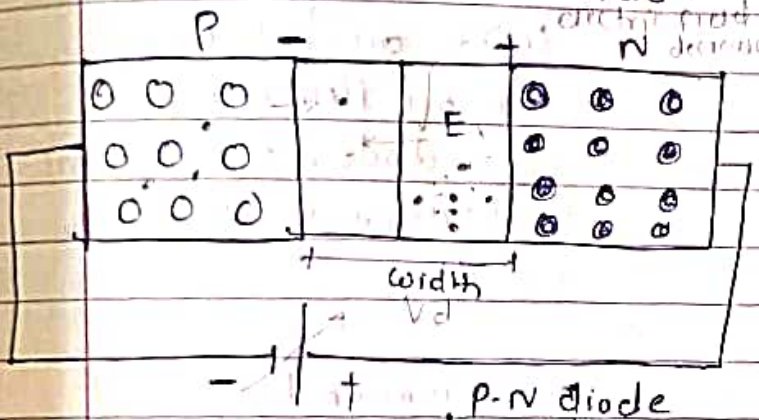


P-N Junction diode

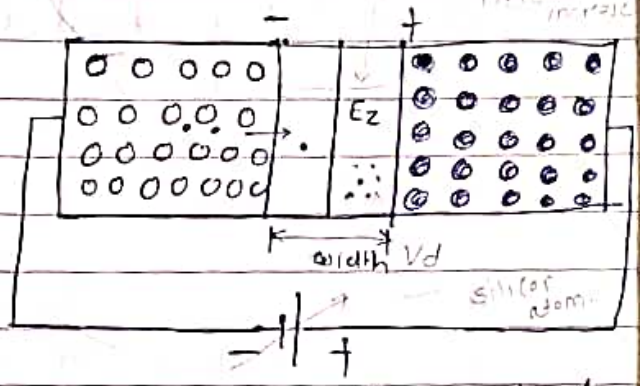


zener diode

> Now we Connect the P-N Junction & zener diode are in reverse bias.



P-N diode fig



Zener diode fig

> the depletion region of Both the diode is increase

> Theory of PN Junction diode	> Theory of zener diode.
> Electric field in a P-N Junction diode is decreasing	Electric field in a zener diode is increases
> when we increases the voltage across diode the minority charge carrier electrons are move in a depletion region.	when we increases the voltage across diode the minority carrier electrons are move in depletion region.
> this charge have a maximum kinetic energy & the presented silicon atom in depletion region to knock out the free electrons	> This free electrons are not collied with the silicon atoms because the depletion region width is thin

Avalanche break down

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> This free electrons atom make the covalent bond of neghbhouring silicon atom & free the electrons in that covalent bond.

> collision are produced in that process & also heat is produced

> so diode is break due to the heat

> That why it does not /or minimum possibility to make collision between the free electrons and silicon atom because The depletion width is thin

> That why heat is not produced in the zener diode.

> & the voltage at certain level are not exceed.

> Electric field is high in zener diode, & this electric field remove the electrons in outside.

>

> But maximum or high current are not produced

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To produced high current the electrons are goes in valance band to conduction band.

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But this process is not done using collision but the this process is done by using high elelectric field.

& electrons are conducting.

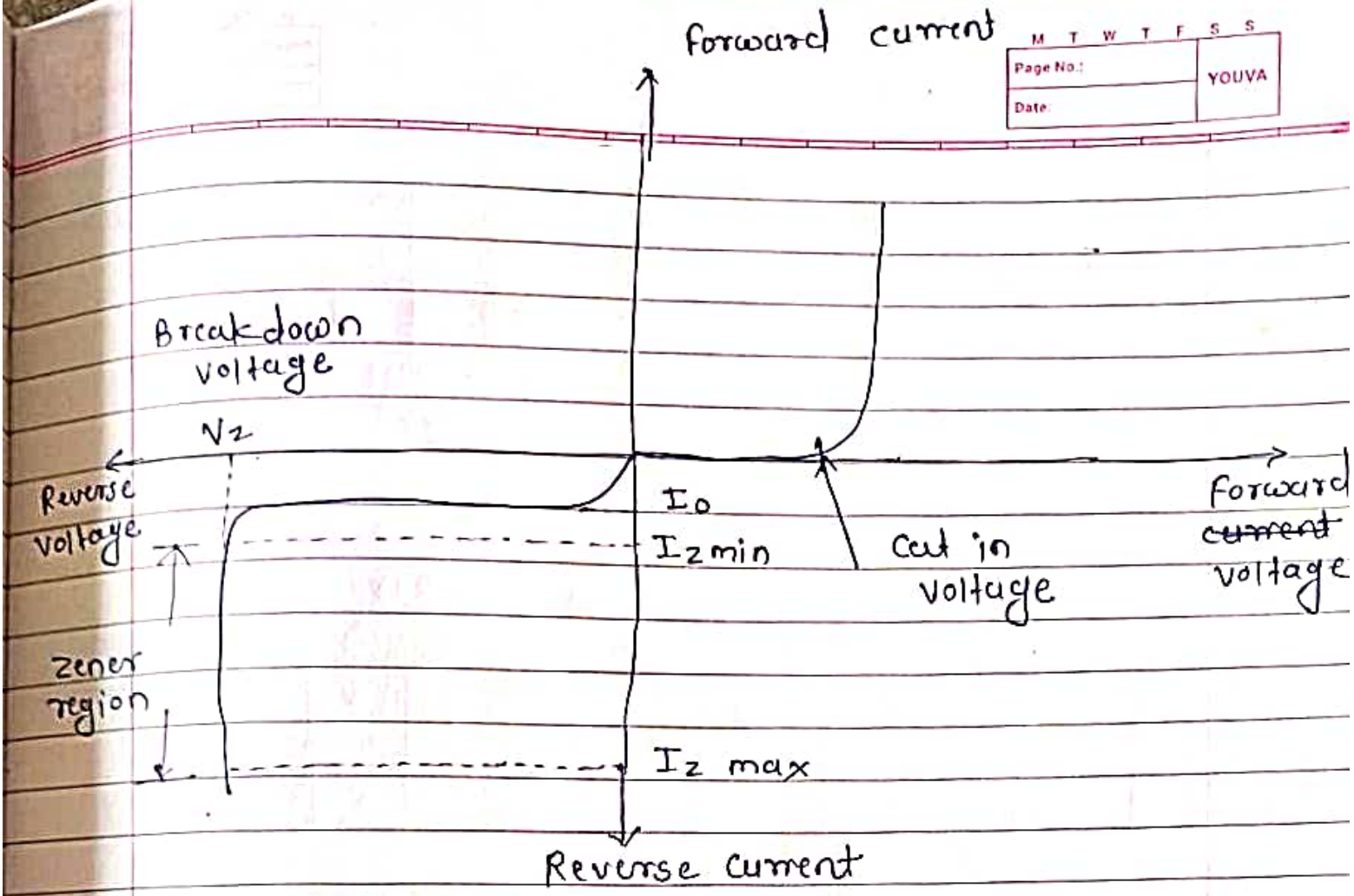


fig > V-I characteristics of a zener diode.

(*) Application of zener diode

① As a voltage reference in emitter follower type voltage regulator
② As a regulated power supply
③ In the protection circuit for MOSFET & opamps.
④ In the clipping circuit
⑤ In the pulse amplifier.

Convert electrical energy into light
LED cnt flow in F.B & it's Block in reverse biase

* Light Emitting Diodes (LED)

> A light emitting diodes (LED) is a P-n junction diode which emits light when electrical energy is applied to it. LED is a two terminal device.
> The terminal are named as anode (A) & the other is as cathode (C)
> A PN Junction formed between the anode & cathode. So LED is basically a PN Junction diode.

Construction of LED

① one of the popular methods of LED construction is to deposite three semiconductor layers on the substrate as shown

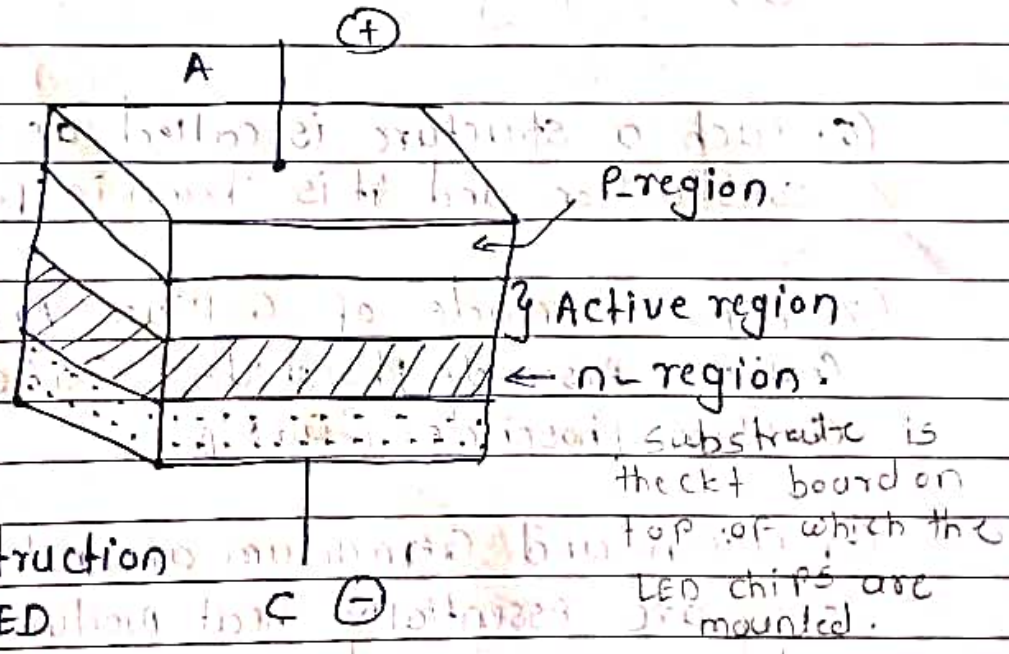


Fig a) Construction of LED

② The Active region exists between the p and n regions.
③ The light emerges from the active side in all the direction when electron hole pair recombine.
④ The disadvantages of this structure is that the LED emits light in all the direction.

⑤ This problem can be solved by placing a small reflective cup inside the basic structure shown in fig b) so as to focus the light in the desired direction.

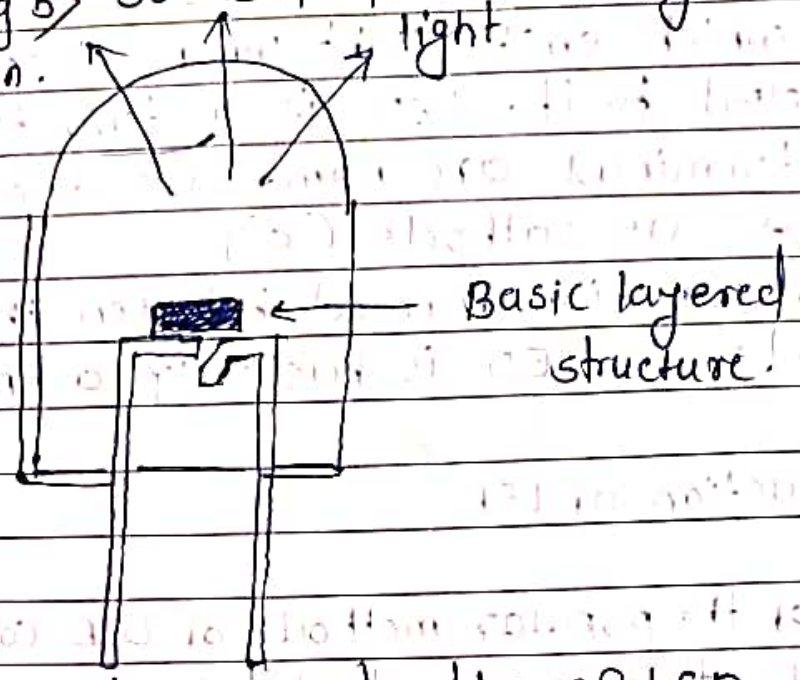


fig b) Cup type Construction of LED

⑥ such a structure is called as a cup type construction and it is shown in fig b)

⑦ LED are made of Gallium Arsenide (GaAs), Gallium Arsenide phosphide (GaAsP) & Gallium phosphide (GaP).

⑧ Silicon and Germanium are not used because they are essentially heat producing material and are very poor in producing light.

Principle of LED operation

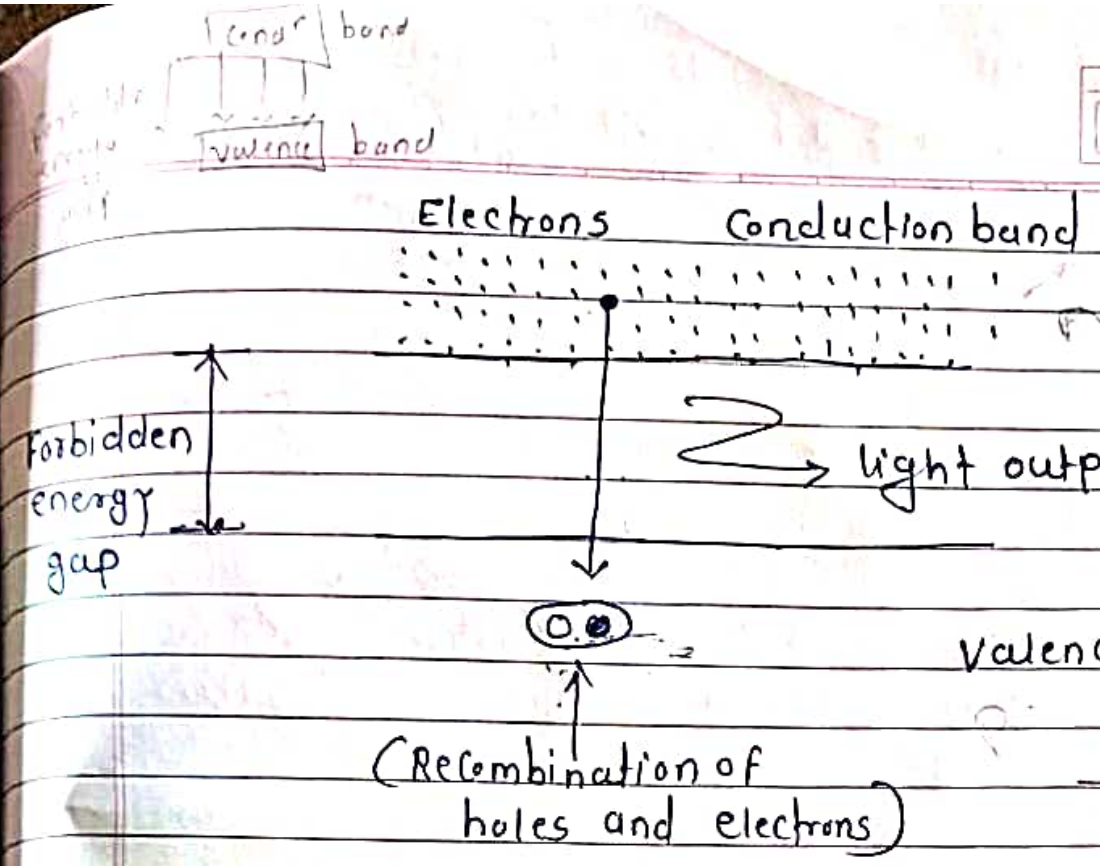


Fig c> Principle of operation of LED

① When the LED is forward biased the electrons in the n-region will cross the junction and recombine with the holes in the P-type material.

② These free electrons reside in the conduction band and hence at a higher energy level than the holes in the valence band.

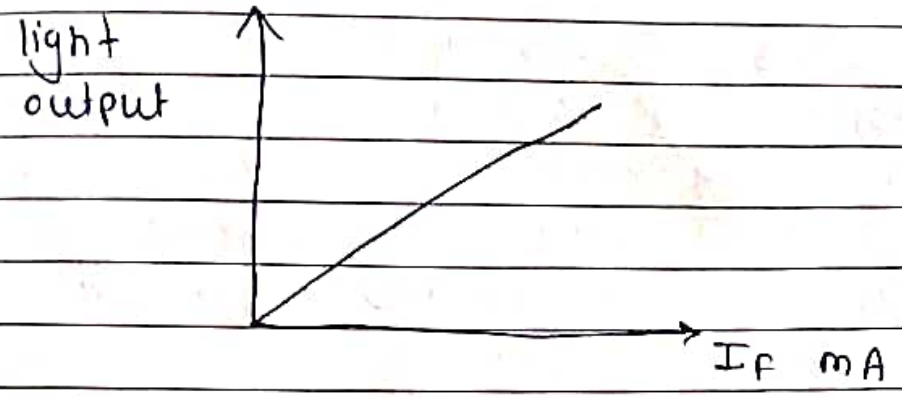
③ when recombination take place, these electrons return back to the valence band which is at a lower energy level than the conduction band.

④ while returning back, the recombining electrons give away the excess energy in the form of light. This shown in Fig c)

⑤ This process is called as electroluminescence.

⑥ In this way an LED emitts light. This is the Principle of operation of LED

output characteristics of LED



Application of LED

1. In the opto coupler
2. In the infrared remote control
3. as indicator in various electronic circuit
4. In seven segment display.